COLLEGE ALGEBRA

Instructor: Engr. Mary Ruth R. Paja

ACTIVITY

(Deadline: 4PM today, 09-03-26)

General Instructions:

- Answer **only** the **odd-numbered** items.
- Use whole sheet of yellow paper and a black ballpoint pen.
- Do not forget to write your name on each answer/work sheet.
- Show your step-by-step solutions.
- Write your work clearly and neatly.

EXERCISES 2.4

In Exercises 1 through 46, find the solution set of the equation.

1. $\sqrt{x} - 5 = 3$
2. $\sqrt{x} + 4 = 7$
3. $\sqrt{x} + 5 = 3$
4. $\sqrt{x} - 4 = 7$
5. $\sqrt{2x - 3} = 2$
6. $\sqrt{2x + 5} = 3$
7. $\sqrt{y} + y = 6$
8. $\sqrt{t} + 6 = t$
9. $\sqrt{3x + 7} = x + 1$
10. $2\sqrt{x + 4} - 1 = x$
11. $\sqrt{x + 5} - \sqrt{x} = 1$
12. $\sqrt{2t + 1} - 2\sqrt{t} - 1 = 0$
13. $\sqrt{3x - 4} + 8 = 0$
14. $\sqrt{5x + 1} + \sqrt{3x} + 1 = 0$
15. $\sqrt{5w + 1} - \sqrt{3w} - 1 = 0$
16. $\sqrt{2 + 4y} + \sqrt{3 - 4y} = 3$
17. $\sqrt{2x + 11} + 1 = \sqrt{5x + 1}$

Answer Key:

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1. $\{64\}$ **3.** $\varnothing$ **5.** $\{\frac{7}{2}\}$ **7.** $\{4\}$ **9.** $\{3\}$ **11.** $\{4\}$
13. $\varnothing$ **15.** $\{0, 3\}$ **17.** $\{7\}$ **19.** $\{-2\}$ **21.** $\{\frac{5}{3}, 3\}$
23. $\{-7\}$ **25.** $\{1, 2\}$ **27.** $\{-3, 0\}$ **29.** $\{\pm 1, \pm 2\}$